

Physical Science Part 3

1. Which of the following BEST describes temperature?
 - A. Temperature measures the total energy in something.
 - B. Temperature measures the kinetic energy in substance.
 - C. Temperature measures the total kinetic energy contained in an object.
 - D. Temperature measures the average molecular kinetic energy in substance.

2. The Earth receives tremendous heat coming from the sun which could be beneficial at one point and harmful to a large extent. Which of the following methods of transferring heat is being illustrated?
 - A. Conduction
 - B. Convection
 - C. Radiation
 - D. Transduction

3. A reaction takes place endothermically. Which of the following statements BEST describes endothermic reactions?
 - A. The temperature of the system drops.
 - B. The temperature of the surrounding rises.
 - C. The temperature of the surrounding drops.
 - D. The temperature of the surrounding remains the same.

4. Which of the following statements on specific heat is TRUE?
 - A. It is an extrinsic property.
 - B. It is always equal to the heat capacity.
 - C. It is the heat capacity per mole of a substance.
 - D. It is equal to zero when the heat capacity is zero.

5. The entropy usually increase when _____.
 - I. A molecule is broken into two or more smaller molecules
 - II. A reaction occurs that results in an increase in the number of moles of gas
 - III. A solid changes to a liquid
 - IV. A liquid changes to a gas
 - A. I and II only
 - B. III and IV only
 - C. I, II and III only
 - D. I, II, III, and IV

6. Change in enthalpy for combustion of fuel is _____.
 - A. negative
 - B. positive
 - C. zero
 - D. may be negative or positive

7. According to the collision theory, when does a chemical reaction occur?
- A. If the colliding molecules have energy equal to greater than the activation energy.
 - B. If the colliding molecules have lesser than the activation energy.
 - C. If the colliding molecules bouncing apart did not change.
 - D. If there is no collision among molecules.
8. Which situation appears to violate the collision theory?
- A. The reaction rate increases as temperature increases.
 - B. The reaction rate decreases as the reaction mixture is cooled.
 - C. The reaction rate increases as the surface area of the reactants increases.
 - D. A reaction rate takes more rapidly when the concentration of the reactants is decreased.
9. Which statements is TRUE about catalysts?
- A. They increase the rate of reaction by increasing the activation energy.
 - B. They provide alternative reaction paths with lower activation energies.
 - C. The amount of catalyst decreases as the reaction progresses.
 - D. They are used up in the reaction.
10. Milk turns sour in one or two days when left at room temperature and yet can remain unspoiled for two weeks when refrigerated. What explains this?
- A. Decrease in temperature speeds up reaction rate.
 - B. Decrease in temperature slows down reaction rate.
 - C. Milk becomes less concentrated when placed in refrigerator.
 - D. Milk becomes more concentrated when placed in refrigerator.
11. What factor affects sawdust to burn faster than a block of wood?
- A. Concentration of reactant
 - B. Presence of catalyst
 - C. Surface area of reactant
 - D. Temperature
12. Which of the following observation is NOT true whenever a system is in chemical equilibrium?
- A. Attainment of equilibrium requires time.
 - B. The rates of opposing reactions are equal.
 - C. The products do not change back to reactants.
 - D. Concentration remain constant at equilibrium even though the opposing reactions are continuing.
13. Which refers to the recorded observations and measurements in an explanation?
- A. Conclusion
 - B. Data
 - C. Graph
 - D. Variables
14. How is hypothesis tested?

- A. By gathering enough information about the problem
- B. By making inferences
- C. By conducting experiment
- D. By formulating conclusion

15. Which summarizes a natural occurrence that may be constantly observed given the same conditions which may be expressed in words or in mathematical form?

- A. Conclusion
- B. Hypothesis
- C. Law
- D. Theory

16. Which of the following is also known as paraffin or saturated hydrocarbon?

- A. Alkane
- B. Alkene
- C. Alkyne
- D. Cycloalkane

17. Which of the following is NOT true about scientific method?

- A. Conclusion is the last step
- B. Scientific method may be applied in every life.
- C. Scientific method is a systematic way of solving problem.
- D. Scientific method is done by following a step-by-step procedure in an exact sequence.

18. Which of the following are properties of organic compound?

- High melting point
- Low boiling point
- High solubility in water
- Non-conductor of electricity

- A. I and II
- B. I and III
- C. II and III
- D. II and IV

19. What is the classification of an organic compound with the formula $\text{CH}_3\text{-O-CH}_3$?

- A. Alcohol
- B. Ether
- C. Phenol
- D. Thiol

20. Which apparatus will you use to measure the mass of an irregular solid correctly?

- A. Empty beaker
- B. Erlenmeyer flash
- C. Graduated cylinder with water
- D. Vernier caliper

21. Which of the following is an organic compound?
- A. Baking soda
 - B. Glucose
 - C. Limestone
 - D. Sulfuric acid
22. Which of the following serves as thermal insular in the body?
- A. Carbohydrates
 - B. Fats
 - C. Nucleic acids
 - D. Proteins
23. In DNA, which is the correct complementary pairing of the nitrogenous bases?
- A. AG and CT
 - B. AA and GG
 - C. AC and GC
 - D. AT and GC
24. Which biomolecule is considered the building blocks of tissues and organs?
- A. Carbohydrates
 - B. Lipid
 - C. Nucleic acid
 - D. Protein
25. What is the main function of the hormone oxytocin in the body?
- A. Controls water secretion by the kidneys
 - B. Stimulates conversion of glycogen to glucose
 - C. Controls blood – glucose level and storage of glycogen
 - D. Stimulates contraction of the uterine muscles and secretion of milk by the mammary glands
26. The half-life of thorium-234 is 24 days. How much 5g of ^{234}Th would be left after 48 days?
- A. 0.625 g
 - B. 1.25 g
 - C. 2.50 g
 - D. 3.75 g
27. What happens when there is a loss of an alpha particle from the nucleus?
- A. No change in the mass number or atomic number
 - B. Loss of 4 in the mass number, loss of 2 in the atomic number
 - C. Loss of 2 in the mass number, loss of 4 in the atomic number
 - D. No change in the mass number, increase of 1 in the atomic number

28. What is the balanced equation of $^{70}_{31}\text{Ga}$ that undergoes beta decay?

- A. $^{70}_{31}\text{Ga} \longrightarrow ^{70}_{31}\text{X} + ^0_0\beta$
- B. $^{70}_{31}\text{Ga} \longrightarrow ^{70}_{32}\text{X} + ^0_{-1}\beta$
- C. $^{70}_{31}\text{Ga} \longrightarrow ^{70}_{31}\text{X} + ^0_{-1}\beta$
- D. $^{70}_{31}\text{Ga} \longrightarrow ^{71}_{31}\text{X} + ^{-1}_0\beta$

29. What is the medical application of the isotope P-32 with a half-life of 4.3 days?

- A. Liver imaging
- B. Gastrointestinal diagnosis
- C. Treatment of leukemia
- D. Treatment of hyperthyroidism

30. Which of the following poses a serious obstacle to the development of commercial fusion power plants?

- A. Scarcity of deuterium
- B. Public concern about safety
- C. Dwindling supply of uranium
- D. Exceedingly high activation energy

31. A radioisotope is considered a tracer when it used to _____.

- A. kill cancerous tissues
- B. Kill bacteria in food
- C. Determine the age of artifacts
- D. Determine the way plants absorb fertilizers

32. Identify the following quantities as scalar or vector respectively: mass of an object, the number of leaves on a tree, and wind velocity.

- A. Scalar, Scalar, Vector
- B. Scalar, Vector, Scalar
- C. Vector, Scalar, Scalar
- D. Vector, Scalar, Vector

33. Which of the following is a scalar quantity?

- A. Acceleration
- B. Temperature
- C. Velocity
- D. Weight

34. Which of the following describes a vector quantity?

- A. It specifies direction only.
- B. It specifies magnitude only.
- C. It can be a dimensionless a vector quantity.
- D. It specifies both magnitude and direction.

35. A man pushes against the wall with 25 pounds of force. The wall does not move. What is the resultant force?

- A. -25 pounds
- B. 0 pounds
- C. 50 pounds
- D. 75 pounds

36. In a coordinate system, a vector is oriented at angle with respect to the x axis. The components of the vector equals the vector's magnitude multiplied by which trigonometric function?

- A. Cos angle
- B. Cot angle
- C. Sin angle
- D. Tan angle

37. An ant on a picnic table travels 30 cm due east, then 25 cm northward and finally 15 cm due west. What is the magnitude of the ant's displacement relative to its point of origin?

- A. 29 cm
- B. 58 cm
- C. 70 cm
- D. 99 cm

38. What will happen to a moving object in the sum of all the forces acting on it is zero?

- A. It will accelerate uniformly.
- B. It will slow down and stop.
- C. It will change the direction of its motion.
- D. It will continue moving with constant velocity.

39. Acceleration is always in the direction of the _____.

- A. displacement
- B. final velocity
- C. initial velocity
- D. net force

40. A brick slides on the horizontal surface. Which of the following will increase the magnitude of the frictional force on it?

- A. Putting a second brick on top
- B. Decreasing the mass of the brick
- C. Increasing the surface area of contact
- D. Decreasing the surface area of contact

Answer Keys:

1. D
2. C
3. C
4. D
5. D
6. A
7. A
8. D
9. A
10. B
11. C
12. C
13. B
14. C
15. C
16. A
17. D
18. D
19. B
20. C
21. B
22. B
23. D
24. D
25. D
26. B
27. B
28. C
29. C
30. D
31. D
32. A
33. B
34. D
35. B
36. C
37. A
38. D
39. D
40. A